

ECO101 - Introduction to Microeconomics

Department of Economics and Social Science (ESS), Brac University

The Production Process: Inputs and Outputs

Lecture: 5

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 10 & 11

Outline

- Distinguish between technological and economic efficiency
- Distinguish between the short run and the long run
- Explain the relationship between a firm's output and labour employed
- Total product, marginal product and average product

Technological and Economic Efficiency

- There are two concepts of production efficiency: technological efficiency and economic efficiency
- **Technological efficiency** occurs when the firm produces a given output by using the least amount of inputs
- **Economic efficiency** occurs when the firm produces a given output at the least cost

Technological and Economic Efficiency

TABLE 10.2 Four Ways of Making 10 TVs a Day

Method		Quantities of inputs	
		Labor	Capital
A	Robot production	1	1,000
B	Production line	10	10
C	Hand-tool production	1,000	1
D	Bench production	100	10

D is technologically inefficient

Technological and Economic Efficiency

TABLE 10.3 The Costs of Different Ways of Making 10 TVs a Day

(b) Wage rate \$150 per day; Capital rental rate \$1 per day

Method	Inputs		Labor cost (\$150 per day)		Capital cost (\$1 per day)		Total cost
	Labor	Capital					
A	1	1,000	\$150	+	\$1,000	=	\$1,150
B	10	10	1,500	+	10	=	1,510
C	1,000	1	150,000	+	1	=	150,001

Capital
intensive

(c) Wage rate \$1 per day; Capital rental rate \$1,000 per day

Method	Inputs		Labor cost (\$1 per day)		Capital cost (\$1,000 per day)		Total cost
	Labor	Capital					
A	1	1,000	\$1	+	\$1,000,000	=	\$1,000,001
B	10	10	10	+	10,000	=	10,010
C	1,000	1	1,000	+	1,000	=	2,000

Labour
intensive

Technological and Economic Efficiency

- Thus, technological efficiency depends only on what is feasible
- Whereas, economic efficiency depends on the relative cost of resources (or inputs)
- Economic efficiency depends on the relative costs of resources
 - > uses a smaller amount of the more expensive resource
 - > a larger amount of the less expensive resource

A firm that is not economically efficient does not maximize profit

Production and Costs

- Firms incur costs when they buy inputs to produce the goods and services that they plan to sell
- **Production function** is the relationship between quantity of inputs used to make a good and the quantity of output of that good
- **Variable inputs** can be varied in the short run
- **Fixed inputs** cannot be varied in the short run

Decision Time Frames

- Firms make many decisions everyday, and all of their decisions are aimed at achieving one overriding goal  maximum attainable profit
- All types of firms in all types of markets make similar decisions about how to produce
- The actions that a firm can take to influence the relationship between output and cost depend on how soon the firm wants to act
- A firm that plans to change its output rate tomorrow has fewer options than one that plans to change its output rate six months or six years from now

Decision Time Frames

To study the relationship between a firm's output decision and its costs, we distinguish between two decision time frames:

- The short run
- The long run

The Short Run

- The **short run** is a time frame in which the quantity of at least one factor of production is fixed
- For most firms, capital, land, and entrepreneurship are fixed factors of production and labour is the variable factor of production
- For a producer of knitwear, the fixed factors of production is its factory building and its knitting machines
- For an electric power utility, the fixed factors of production is its buildings, generators, computers, and control systems

The Short Run

- To increase output in the short run, a firm must increase the quantity of a variable factor of production, which is usually labour. So to produce more output, a producer of knitwear must hire more labour and operate its knitting machines for more hours a day
- Similarly, an electric power utility must hire more labour and operate its generators for more hours a day
- In the short-run a firm can increase or decrease its output by increasing or decreasing the amount of labour it hires

The Long Run

- The **long run** is a time frame in which the quantities of all factors of production can be varied
- To increase output in the long run, a firm can change its factors of production that were fixed in the short run as well as the quantity of labour it hires
- For example a knitwear producer can decide whether to install more knitting machines, use a new type of machine, reorganize its management, or hire more labour

Short-Run Technology Constraint

- To increase output in the short run, a firm must increase the quantity of labour employed
- The relationship between output and the quantity of labour employed is described by using three related concepts :

1. Total product

2. Marginal product

3. Average product

Total, Marginal & Average Products

- **Total product** is the maximum output that a given quantity of labour can produce
 - As a firm employs more labour, total product increases
- The **marginal product** of labour is the increase in total product that results from a one-unit increase in the quantity of labour employed, with all other inputs remaining the same
- **Average product** tells how productive workers are on average. The average product of labour is equal to total product divided by the quantity of labour employed

Product Schedules

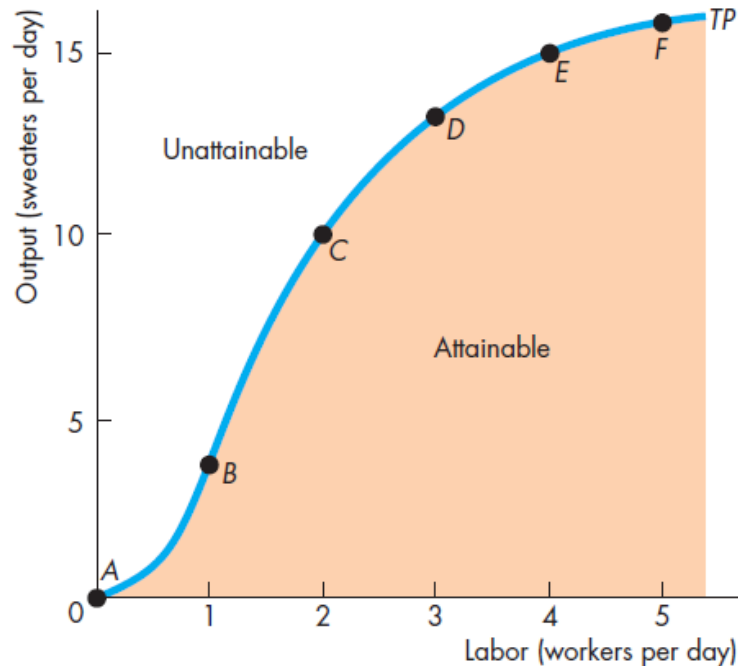
	Labor (workers per day)	Total product (sweaters per day)	Marginal product (sweaters per additional worker)	Average product (sweaters per worker)
A	0	0	4	
B	1	4	6	4.00
C	2	10	3	5.00
D	3	13	2	4.33
E	4	15	1	3.75
F	5	16		3.20

Product Curves

- The product curves are graphs of the relationships between employment and the total product, marginal product and the average product
- They show how total product, marginal product, and average product change as employment changes
- They also show the relationships among these three concepts

Total Product Curve

FIGURE 11.1 Total Product Curve



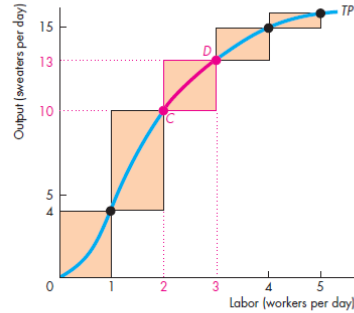
- TP curve is a graph of the total product schedule
- All the points that lie above the curve are unattainable
- Points that lie below the curve, in the orange area, are attainable, but they are inefficient—they use more labour than is necessary to produce a given output
- Only the points on the total product curve are technologically efficient

Marginal Product Curve

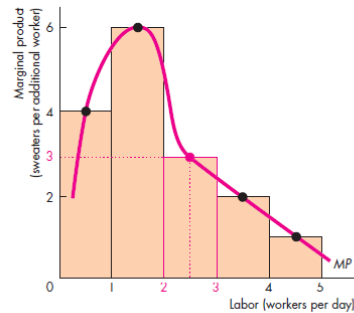
- **Marginal product** is also measured by the slope of the total product curve
- Slope of a curve is the change in the value of the variable measured on the y-axis (output) divided by the change in the variable measured on the x-axis (labour) as we move along the curve
- A one-unit increase in labour, from 2 to 3 workers, increases output from 10 to 13 sweaters, so the slope from point C to point D is 3 sweaters per additional worker, the same as the marginal product
- Varying the amount of labour in the smallest units possible, we can draw the marginal product curve
- The height of this curve measures the slope of the total product curve at a point

Marginal Product Curve

FIGURE 11.2 Total Product and Marginal Product



(a) Total product



(b) Marginal product

- The steeper the slope of the total product curve (TP) in part (a), the larger is marginal product (MP) in part (b)
- Marginal product increases to a maximum (in this example when 1.5 workers a day are employed) and then declines due to diminishing marginal product

Marginal Returns

The shapes of the product curves are similar because almost every production process has two features:

- Increasing marginal returns initially
- Diminishing marginal returns eventually

Increasing Marginal Returns

- **Increasing marginal returns** occur when the marginal product of an additional worker exceeds the marginal product of the previous worker
- Increasing marginal returns arise from increased specialization and division of labour in the production process
- Example: if a knitting firm employs one worker, then that person must learn all the aspects of sweater production: running the knitting machines, fixing breakdowns, packaging and mailing sweaters, buying and checking the type and color of the wool. All these tasks must be performed by that one person
- Now if the firm hires a second worker, then the two workers can specialize in different parts of the production process and can produce more than twice as much as one worker. The marginal product of the second worker is greater than the marginal product of the first worker. Marginal returns are increasing

Diminishing Marginal Returns

- **Diminishing marginal returns** occur when the marginal product of an additional worker is less than the marginal product of the previous worker
- Diminishing marginal returns arise from the fact that more and more workers are using the same capital and working in the same space
- Example: if the knitting firm hires a third worker, output increases but not by as much as it did when it hired the second worker because by hiring a third worker, the factory produces more sweaters, but the equipment is being operated closer to its limits
- Hiring more and more workers continues to increase output but by successively smaller amounts. Marginal returns are diminishing
- This phenomenon is called — the law of diminishing returns

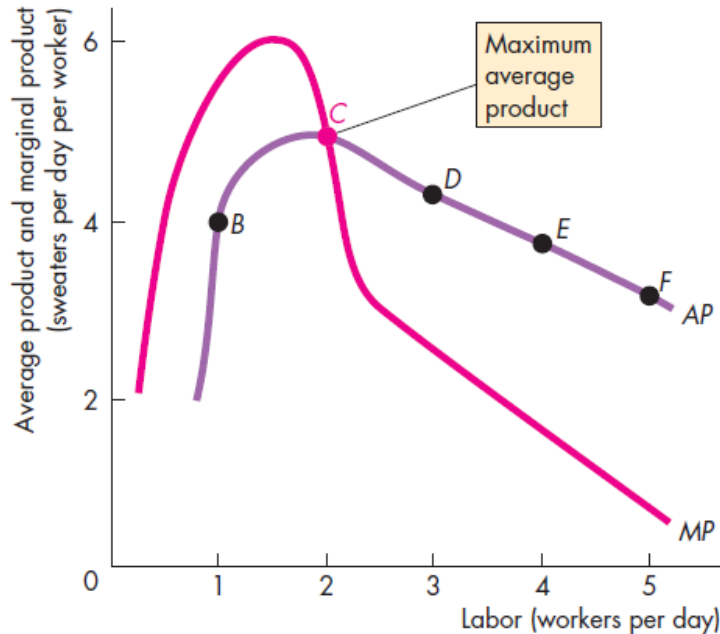
The Law of Diminishing Returns

The **law of diminishing returns** states that,

As a firm uses more of a variable factor of production with a given quantity of the fixed factor of production, the marginal product of the variable factor eventually diminishes

Average Product Curve

FIGURE 11.3 Average Product



- AP curve corresponds to those same rows in product schedule table
- With 1 worker, marginal product exceeds average product, so average product is increasing
- With 2 workers, marginal product equals average product, so average product is at its maximum
- With more than 2 workers, marginal product is less than average product, so average product is decreasing